TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	90.0 ± 0.77
Correlation Stability	95.0 ± 0.77
Distribution Consistency	91.6 ± 0.49
Entropy Preservation	82.4 ± 0.49
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.1 ± 0.30
Data Density	100.0 ± 0.00
Coverage Score	87.4 ± 3.88
Feature Accuracy	96.5 ± 1.02
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.005	+0.160	4.2%	OK
Glucose	120.895	119.674	-1.220	1.0%	OK
BloodPressure	69.105	69.792	+0.686	1.0%	OK
SkinThickness	20.536	21.181	+0.644	3.1%	OK
Insulin	79.799	82.352	+2.552	3.2%	OK
BMI	31.993	32.148	+0.156	0.5%	OK
DiabetesPedigreeFunction	0.472	0.495	+0.023	4.9%	OK
Age	33.241	33.077	-0.164	0.5%	OK
Outcome	0.349	0.319	-0.030	8.5%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	90.0 ± 0.77
Correlation Stability (*2)	95.0 ± 0.77
Distribution Consistency (*3)	91.6 ± 0.49
Entropy Preservation (*4)	82.4 ± 0.49
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.1 ± 0.30
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	87.4 ± 3.88
Feature Accuracy (*10)	96.5 ± 1.02

Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7467532467532467	0.7344155844155844	0.7454545454545455	-0.0012987012987012991	
F1	0.6166009236312779	0.5562997784100342	0.5996832361677265	-0.016917687463551507	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	0.7201298701298702	0.7266233766233767	-0.002597402597402593	
F1	0.592152063889281	0.553092785436399	0.5801904451394191	-0.011961618749861912	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7162337662337661	0.6857142857142857	0.7097402597402597	-0.006493506493506484	
F1	0.5770339369355421	0.5077639275667112	0.5400533603823078	-0.03698057655323426	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.7474025974025974	0.7584415584415585	-0.005194805194805208	
F1	0.6399852354635025	0.5895941287112556	0.6174744102631189	-0.022510825200383 49	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.7149350649350657	0.7597402597402597	0.0032467532467532314	
F1	0.5733680734057942	0.3848466122847812 3	0.5654617899430565	-0.007906283462737774	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7298701298701299	0.7616883116883116	0.012337662337662347	
F1	0.6076604413342895	0.5429565709499797	0.6190238098905021	0.011363368556212583	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7571428571428571	0.7383116883116883	0.7597402597402596	0.002597402597402695	
F1	0.6193219200525574	0.5429982026494407	0.6093801500479936	-0.009941770004563855	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	0.7564935064935064	0.7668831168831168	-0.005194805194805195	
F1	0.6262914445612621	0.570274747952019	0.6082640508130468	-0.018027393748215336	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6766233766233766	0.6610389610389611	0.6909090909090908	0.014285714285714285	
F1	0.5115304626989783	0.3760790752840224	0.48290871229049265	-0.028621750408485656	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.7201298701298702	0.7331168831168832	-0.021428571428571443	*
F1	0.5998259463389416	0.4867609556817029 7	0.5616039105106021	-0.038222035828339405	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	118.0	77.0	26.0	29.0	31.791264733083 18	0.6988302523775 066	14.0	1.0	1.0
2	1.0	102.0	60.0	18.0	20.0	31.456068167083 146	0.2175738724333 121	22.0	0.0	1.0
3	-1.0	134.0	11.0	-11.0	45.0	36.4340462450944	0.0521767762753 29684	26.0	1.0	1.0
4	2.0	165.0	75.0	33.0	164.0	40.712212319604 44	0.6588599257274 974	28.0	1.0	1.0
5	5.0	118.0	79.0	-14.0	6.0	31.418285703462 91	0.1472318651431 4257	65.0	0.0	1.0
6	1.0	91.0	59.0	13.0	18.0	28.027158187303 417	0.3720924040212 0586	32.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	5.0	181.0	82.0	20.0	692.0	29.793504129196 105	0.2419818245479 8628	62.0	0.0	2.0
2	1.0	59.0	35.0	12.0	82.0	25.399073561820 405	0.3178713974872 043	21.0	0.0	2.0
3	7.0	180.0	34.0	16.0	-4.0	33.477578763170 54	1.9995582709326 223	28.0	1.0	2.0
4	11.0	139.0	87.0	42.0	278.0	25.466659493095 385	0.2161028026239 4596	45.0	1.0	2.0
5	9.0	105.0	62.0	42.0	60.0	36.4177374949334	0.3735421764240 014	48.0	0.0	2.0
6	5.0	57.0	65.0	25.0	314.0	33.155562238617 38	0.4731235579226 9684	55.0	0.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	80.0	49.0	20.0	78.0	23.061012498063985	0.790115520416762	21.0	0.0	3.0
2	9.0	88.0	80.0	0.0	0.0	38.902289308300475	0.40941174540438985	58.0	0.0	3.0
3	6.0	64.0	107.0	45.0	5.0	45.640178355824601	0.29059494011710046	41.0	0.0	3.0
4	8.0	99.0	23.0	-5.0	0.0	33.585735679835303	0.6105290553936575	32.0	0.0	3.0
5	1.0	87.0	65.0	27.0	25.0	25.015898211219763	0.5398592526525879	26.0	0.0	3.0
6	3.0	108.0	80.0	39.0	0.0	30.265156811515903	0.9960136976463905	31.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	9.0	112.0	86.0	0.0	0.0	47.439821651711204	0.5038369285450849	53.0	1.0	4.0
2	11.0	137.0	99.0	33.0	378.0	42.619526785708706	0.5135121250353493	41.0	1.0	4.0
3	3.0	78.0	63.0	-5.0	97.0	28.750544844984084	0.2672040489141754	27.0	0.0	4.0
4	7.0	100.0	65.0	16.0	0.0	22.079590555835807	0.2034621512913771	33.0	0.0	4.0
5	-2.0	158.0	96.0	31.0	90.0	44.652802011731504	0.8616061387332481	2.0	0.0	4.0
6	2.0	101.0	64.0	36.0	72.0	10.726369232823075	-0.14668142284060426	25.0	0.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	133.0	77.0	10.0	49.0	22.411289076951746	0.2856910130190055	23.0	0.0	5.0
2	5.0	74.0	75.0	-1.0	10.0	39.840101061641036	0.39489189636475747	36.0	0.0	5.0
3	6.0	167.0	93.0	38.0	74.0	32.6582914031066	0.42304451638168605	72.0	1.0	5.0
4	4.0	92.0	61.0	2.0	-10.0	23.442881996731536	0.2583044144412532	21.0	0.0	5.0
5	7.0	93.0	62.0	0.0	0.0	26.968344824432 1	0.5732031152270759	25.0	0.0	5.0
6	2.0	134.0	67.0	18.0	136.0	31.637144259418 53	0.4458340209289323	26.0	0.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	112.0	70.0	14.0	-23.0	23.016393166958142	0.4271659326785946	24.0	0.0	6.0
2	-3.0	105.0	81.0	45.0	88.0	45.609737345375954	0.9469785519700031	24.0	1.0	6.0
3	1.0	143.0	85.0	46.0	364.0	34.446813674721 79	0.6050946162101257	23.0	1.0	6.0
4	9.0	189.0	83.0	0.0	0.0	33.879521788799 61	0.4100453236287 9237	25.0	1.0	6.0
5	4.0	92.0	60.0	29.0	95.0	30.442160666816573	0.2847179112615 6154	31.0	0.0	6.0
6	1.0	161.0	60.0	1.0	58.0	24.199369563992285	0.3967149023191 6995	45.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	159.0	60.0	21.0	449.0	28.779351371019857	0.7414154993280937	26.0	0.0	7.0
2	8.0	125.0	85.0	26.0	230.0	33.646551464626555	0.5892140829753882	53.0	1.0	7.0
3	8.0	128.0	77.0	13.0	145.0	42.826026717694084	0.2888193534064097	14.0	1.0	7.0
4	1.0	101.0	62.0	22.0	117.0	26.358540308248833	0.2086325248202836	23.0	0.0	7.0
5	8.0	157.0	86.0	29.0	581.0	39.832449642116941	0.7345468712009563	55.0	1.0	7.0
6	-3.0	130.0	76.0	14.0	101.0	37.725694851669196	-0.13928686350257047	56.0	0.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	92.0	0.0	-4.0	0.0	28.557933132066154	0.4179418635659723	25.0	0.0	8.0
2	4.0	105.0	82.0	27.0	87.0	32.856017122802766	0.4377043007428687	38.0	0.0	8.0
3	5.0	87.0	67.0	43.0	381.0	41.767487664904577	0.7668765984485414	34.0	1.0	8.0
4	1.0	164.0	92.0	33.0	492.0	38.046128063201166	0.3336041486164856	25.0	0.0	8.0
5	-0.0	115.0	101.0	34.0	50.0	29.373846659617865	0.4310458820178829 6	24.0	0.0	8.0
6	2.0	83.0	66.0	17.0	-18.0	19.731931779819696	0.4808480259086902	38.0	0.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	7.0	179.0	98.0	20.0	-2.0	25.868540951389107	0.24123642123924285	47.0	1.0	9.0
2	1.0	81.0	68.0	3.0	177.0	25.964983808642002	0.2361185701735021	-15.0	0.0	9.0
3	0.0	185.0	74.0	28.0	232.0	43.940687706382974	0.98943254541590 1	22.0	0.0	9.0
4	5.0	107.0	72.0	1.0	-50.0	36.8392698918868 2	0.17716650260005 75	32.0	0.0	9.0
5	1.0	143.0	72.0	24.0	144.0	32.863750877394 41	0.43601235340584 7	25.0	0.0	9.0
6	1.0	111.0	85.0	23.0	98.0	33.547909717690196	0.61149025684092 35	27.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	154.0	83.0	9.0	153.0	31.870136825728 7	1.24584193944377 12	19.0	1.0	10.0
2	14.0	143.0	106.0	29.0	51.0	35.980211074324 245	0.42131351190491 85	60.0	1.0	10.0
3	7.0	122.0	60.0	31.0	267.0	41.111743956765 004	0.14306894945216 875	23.0	0.0	10.0
4	2.0	118.0	59.0	17.0	115.0	39.309049243443 57	0.62053847127594 3	30.0	0.0	10.0
5	7.0	108.0	81.0	12.0	0.0	31.325079222596 62	0.79119718973327 19	45.0	1.0	10.0
6	0.0	135.0	17.0	-1.0	0.0	30.258267064271 18	0.60851847191203 57	30.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan